

MATH Unit 3 — Probability & Statistics Complete Notes

Mathematics (MATH) · Unit 3 · Probability & Statistics · 2024

Overview

These notes cover **Probability & Statistics**. Engineering and discrete mathematics — linear algebra, calculus, probability, logic and graph theory for computer science. Each section states the core idea, a worked intuition and the points examiners look for.

1. Conditional Probability

$P(A|B) = P(A \text{ and } B) / P(B)$ gives the probability of A given B occurred.

Why it matters: Conditional Probability appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of conditional probability in one line.
- A small example showing conditional probability in practice.
- Common mistakes students make and how to avoid them.

2. Bayes' Theorem

Bayes' theorem updates prior probabilities using observed evidence.

Why it matters: Bayes' Theorem appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of bayes' theorem in one line.
- A small example showing bayes' theorem in practice.
- Common mistakes students make and how to avoid them.

3. Random Variables

Discrete and continuous random variables are described by PMFs, PDFs and CDFs.

Why it matters: Random Variables appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of random variables in one line.
- A small example showing random variables in practice.
- Common mistakes students make and how to avoid them.

4. Binomial and Poisson Distributions

Binomial models successes in n trials; Poisson approximates rare events.

Why it matters: Binomial and Poisson Distributions appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of binomial and poisson distributions in one line.
- A small example showing binomial and poisson distributions in practice.
- Common mistakes students make and how to avoid them.

5. Normal Distribution

The normal distribution is symmetric about its mean; z-scores standardise values.

Why it matters: Normal Distribution appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of normal distribution in one line.
- A small example showing normal distribution in practice.
- Common mistakes students make and how to avoid them.

6. Correlation and Regression

Karl Pearson's coefficient measures linear correlation; regression lines predict one variable from another.

Why it matters: Correlation and Regression appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of correlation and regression in one line.
- A small example showing correlation and regression in practice.
- Common mistakes students make and how to avoid them.

Quick revision

Conditional Probability · Bayes' Theorem · Random Variables · Binomial and Poisson Distributions · Normal Distribution · Correlation and Regression